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# 14. Worked example: IVP and the matrix exponential

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 Calculator

Using the matrix exponential to solve a system



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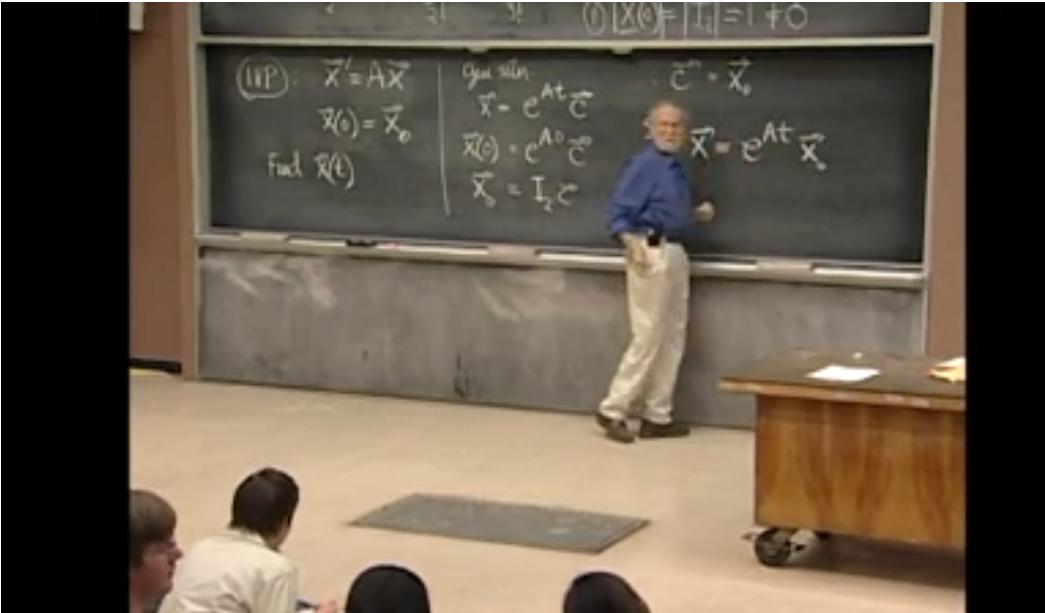
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e to the t plus e  
to the negative t divided by 2.  
And the same similar thing for sinh t.  
So there is your first explicit exponential  
matrix,  
calculated according to the definition.  
And what we have found is the solution  
to the system x  
prime equals y, y prime equals x--  
a fundamental matrix, in other words,  
**are cosh t and sinh t satisfy both  
solutions to that system.**

Initial value problem



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also  
a column vector.  
You cannot multiply a column vector out  
front and the square  
matrix afterwards.  
You can't.  
If you want to multiply a matrix by a  
column vector,  
it has to come afterwards so you can do  
zing zing.  
There is no zing.  
**You see.**  
You can't put it in front.  
It doesn't work.  
So it must go behind.  
That's the only place you might get  
tripped up.  
And as I say, if you try to type that in  
using MATLAB,  
you will immediately get error

The solution to initial value problem

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \quad \mathbf{x}(0) = \mathbf{x}_0$$

(4.171)

can be written nicely in terms of the matrix exponential solution, namely,

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0.$$

(4.172)

14. Worked example: IVP and the matrix exponential

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